

SpaceTree User Guide

Version 0.1.0 for macOS 12 and later. A viewer for allocated disk use. It does not change files.

SpaceTree is a native Mac window that scans a folder, or the whole startup disk, and shows where the allocated bytes went. The top of the window is a file list and an extension legend. The bottom is a treemap of those same bytes. Bundle id `com.cgfixit.spacetree`. It is not affiliated with TreeSize or WinDirStat.

Open the app

The ship bundle is `dist/SpaceTree.app` inside the SpaceTree repo, built by `./scripts/make-app.sh`. The bundle is ad-hoc signed, not an Apple Developer ID signature, so Gatekeeper can block the first open.

In Finder, right-click `SpaceTree.app` and choose Open, then confirm the dialog. If macOS still refuses, open System Settings, Privacy and Security, allow the blocked app, and open it again. After that, a normal double-click is enough. Full Disk Access is optional. Without it, some protected system paths are left out of the tree.

The window opens at 1200 by 780 points, and will not shrink below 800 by 500. Quit with File, Quit or Command-Q. If you rebuild the app, quit fully before opening it again. The open command reuses a process that is already running, so a rebuild does not show up until that process is gone.

```
open /Users/cg/Documents/GitHub/spacetree/dist/SpaceTree.app
```

Scan a folder

The top bar is the scan control: a path field, Browse..., and Scan. The path field starts as your home folder. Type or paste any folder, including / for the startup disk. Browse... opens a folder picker and writes the chosen path into the field.

Click Scan, or press Enter, to start. Command-Enter does not start a scan. Scan is disabled while a walk is already running. An empty path fails with the message path is empty. There is no progress percent and no cancel in this version. The center of the window and the status bar both say Scanning... plus the path until the tree is ready. Large trees, such as a home folder or /, take longer. Wait for the table.

When the scan finishes, the first row is the folder you scanned, and that row starts expanded. A new scan clears the selection and zooms the map back out to the whole scan.

Window layout

The list and the legend share the top pane. That pane starts at about 42 percent of the window height. Drag the bar between the list and the map. The split stays between 20 and 80 percent.

The file list is on the left. The Extension column is on the right, about 180 points wide (at least 140, and at most 28 percent of the width). Its vertical scrollbar stays visible. Each legend row is a color swatch, a label such as `.png` or `(none)`, and a short description such as PNG Image or No Extension. jpeg, tiff, yml, and htm are folded into jpg, tif, yaml, and html. Rows are ordered by allocated bytes, largest first. The legend does not show a size number. The same colors paint the proportion bars and the treemap.

File list

Horizontal and vertical scrollbars stay visible. You can drag column edges. The name column starts at 280 points and clips long names instead of widening the window.

Column	What it shows
Name	Folder or file name. A triangle expands or collapses a folder. Click the triangle to toggle. Files have no triangle. Indentation shows nesting. A symlink is dim and reads name -> target. That row is the link, not a second copy of the target.

Column	What it shows
Size Proportion	A bar of this row against the folder you scanned, colored by extension. The header sorts by physical size, the same key as Physical Size.
Percentage	This row's share of the scanned tree, to two decimal places. It is not the volume percent in the status bar. The header also sorts by physical size.
Physical Size	Allocated bytes on disk, in 1024-based units (B, KB, MB, GB, TB), one decimal place above 1024 bytes. This is the default sort, largest first. Hover the number for the exact byte count.
Logical Size	Sum of file lengths. Sparse files and clones can be larger here than in Physical Size. Its own sort. Hover for the exact byte count.
Files	Number of file leaves under this row, including clones whose allocated size is 0. Its own sort.

Click a column header to sort. Click the same header again to reverse direction. A mark on the header shows the active column. Name starts A to Z. Physical Size, Logical Size, and Files start largest first. Size Proportion and Percentage follow the physical-size sort, so they do not get their own mark. Equal values break alphabetically by name.

Date modified and date created are not columns in the window. Those times appear only in the command-line text dump.

Select, zoom, and Finder

A click highlights a row. It does not move the map. After a fresh scan the map is the whole scan, so the highlight sits inside that full view.

Action	What it does
Click a row	Highlight that file or folder. The map stays at its current zoom.
Double-click a folder row	Highlight the folder and zoom the map into it.
Double-click a file row	Highlight the file and reveal it in Finder.
Right-click a row	Reveal in Finder, or Copy Path.
File menu, or Command-O	Reveal in Finder. The menu item and the shortcut work only when a row is selected.
File menu, or Command-C	Copy the full path. Ignored for the row while the path field is focused, so you can still copy text from that field.
Command-Q	Quit SpaceTree.

Reveal runs `open -R --` and then the path, so a name that starts with a dash is still a path. After Reveal or Copy, the status bar says revealed ... or copied If Finder cannot open the path, the bar shows the error instead.

Status bar

The bottom bar is the summary of the last finished scan, plus the selection.

Readout	What it means
Ready	Shown before the first scan.
Scanning...	A walk is in progress. The path is included. The center of the window says the same thing.
Red text	The scan failed. The message includes the path and the error.
walkable	Allocated size of the scanned tree, after clone sharing and skipped mounts. This is the folder total, not Disk Utility's used figure. Hover for the exact byte count.

Readout	What it means
volume	Total capacity of the volume that contains the scan path. Hover for the exact byte count.
% figure	The third number after walkable and volume. It has no word in front of it. It is walkable size divided by volume capacity, the same share as the root row against the volume. A normal value shows four decimal places. A tiny value uses scientific notation. Zero shows 0%.
selected	Allocated size of the highlighted row, then its full path. Hover the size for the exact byte count.

Treemap

The map draws allocated bytes. Color matches the extension legend. It opens on the whole scan, not on a zoomed folder.

Action	What it does
Click a tile	Select that item and expand its parent folders in the list.
Double-click a folder tile	Zoom the map into that folder.
Double-click a file tile	Reveal that file in Finder.
Zoom out	Shown at the top left only while you are zoomed in. Each click steps to the parent folder, then back to the scan root.

The selected tile has a white outline. Tiles large enough for shading get a lighter center. A folder thinner than 4 points is drawn as one block instead of its children.

On a large scan, one region can hold more items than the map will draw as separate tiles. The budget is about one tile per 16 pixels, kept between 64 and 1024. The largest items stay as their own tiles. The rest fold into one gray hatched tile labeled Other. That tile is the merged remainder, not empty space. A small folder such as Desktop usually stays under the cap, so each file keeps its own tile. Double-click does not zoom into Other.

Why the numbers disagree

You do not need this section to scan. It explains why SpaceTree, Disk Utility, and du can print different totals for the same disk.

Physical Size and walkable are allocated bytes: the blocks the file occupies, times 512. A folder's size is the sum of its children, not the size of the folder inode. Logical Size is the sum of file lengths, which is what Finder often shows in Get Info. Sparse files and clones can therefore look larger in Logical Size than in Physical Size.

The Percentage column is that row's share of the folder you scanned. The percent in the status bar is the walkable total divided by the volume's capacity. Those are different denominators.

APFS clones share extents. SpaceTree counts each clone id once, so a folder of duplicated files is not billed twice. Later copies can still appear as rows with 0 B, and they still count in Files. A symlink to a directory is followed only when it is the path you chose to scan, which is why /tmp can be a tree. Child symlink directories stay as leaves and are not walked.

/System/Volumes is skipped unless that path is the scan root. A scan of / does not count /Users a second time through the Data volume. walkable can sit well below the used figure in Disk Utility. That gap is expected. The walk does not include every protected or unmounted volume, and it is not a bug to close by scanning those volumes from /.

Command line

The same binary inside SpaceTree.app runs from Terminal. From the repo you can use cargo run. The bundled executable is Contents/MacOS/spacetreemap.

Command	What it does
<code>spacetre</code>	Open the window. The path field is your home folder. Nothing is scanned until you click Scan.
<code>spacetre --scan <path></code>	Print the tree to stdout and exit. Does not open the window.
<code>spacetre --gui-scan <path></code>	Open the window and start a scan of that path immediately.
<code>spacetre --help</code>	Print the short usage lines to stderr. -h does the same.

The text dump starts with `volume_total_bytes` and `root_size_bytes`. Each later line is indented by depth and contains allocated bytes, percent of volume, modified time as Unix seconds, created time as Unix seconds, dir or file, and the name. A missing time prints 0.

```
cargo run -- --scan ~/Documents
dist/SpaceTree.app/Contents/MacOS/spacetre --scan /
dist/SpaceTree.app/Contents/MacOS/spacetre --gui-scan ~
```

Rebuild after code changes with `./scripts/make-app.sh` from the repo root, quit SpaceTree, then open the new bundle.

A short workflow

1. Open SpaceTree.app. Right-click Open the first time.
2. Browse to the folder you care about, or type `/`.
3. Click Scan and wait until the table appears. The map shows the whole scan.
4. Click Physical Size if the list is not already largest-first.
5. Expand the biggest folders until you see the files that dominate the total.
6. Click a row to highlight it inside the full map. Double-click a folder to zoom the map. Zoom out steps back to the parent, then to the scan root.
7. Select a row and press Command-O to jump to Finder, or Command-C to copy the path.
8. Compare walkable with volume. If walkable is less than Disk Utility used, that is expected.

SpaceTree 0.1.0. macOS 12 or later. The window is the file list, the extension legend, and the treemap described above.